

Base Table Creation Documentation

1. Objective

The goal is to create a **customer-level base table** for 1996, combining client, account, transaction, order, and district data, enriched with engineered features. The final table is designed for predicting **loan and credit card uptake in 1997**.

2. Data Sources

Dataset	Description
client	Contains static client information, including birth number and demographics.
account	Account-level information, including account opening date and transaction frequency.
card	Details of credit cards issued to clients.
district	Demographic and socio-economic characteristics of districts.
disp	Relationship between clients and accounts (owners or co-owners).
order	Payment orders (debits) on accounts, including type of payment.
trans	All transactions on accounts, including type, operation, and category.
loan	Loans granted to accounts.

3. Data Preparation Steps

3.1 Client Features

- **Birth year, month, day** extracted from birth_number.
 - **Gender** determined by birth month (months > 50 → female).
 - **Age** computed as 1999 - birth_year.
 - **Age group** calculated as age // 10 * 10.
 - Verified distributions by age group and gender.
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3.2 Account Features

- **Length of relationship (lor)** = 1996 - account opening year.
 - **Transaction frequency** translated to English (MONTHLY, WEEKLY, TRANSACTION).
 - Missing values handled and distribution verified.
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3.3 DISP Features

- Selected **owners only** (type=='OWNER') to ensure primary account access.
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3.4 Order Features

- Translated k_symbol to English categories:
 - POJISTNE → Insurance payment
 - SIPO → Household
 - LEASING → Leasing
 - UVER → Loan payment
 - Aggregated per account:
 - num_orders_1996: total number of orders.
 - total_order_amount_1996: total amount of orders.
 - Binary indicators for presence of each order type.
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3.5 Transaction Features

- Translated:
 - type: PRIJEM → Credit, VYDAJ, VYBER → Withdrawal
 - operation: Credit card withdrawal, Cash in, etc.
 - k_symbol: Payment types like Insurance, Household, Loan.

- Aggregated per account:
 - **Total credit and total withdrawal**
 - **Counts** of operations and payment categories
 - **Balance statistics:** mean and standard deviation
 - **RFM Features:**
 - frequency: number of transactions
 - monetary: total transaction amount
 - recency: days since last transaction relative to Dec 31, 1996
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3.6 District Features

- Merged district socio-economic characteristics (A2–A16) with client data:
 - Population, municipalities, urban ratio, salary, unemployment rates, entrepreneurs, crimes, etc.
 - Missing values replaced with 0.
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3.7 Target Variables

- **Loan granted in 1997 (loan_1997):** 1 if the account received a loan, else 0.
 - **Credit card granted in 1997 (card_1997):** 1 if the client received a credit card, else 0.
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4. Feature Engineering

4.1 Transaction Ratios

Feature	Description
credit_to_withdrawal_ratio	Total credit / total withdrawal
net_cash_flow	Total credit – total withdrawal
net_cash_flow_ratio	Net cash flow / total transactions
credit_to_balance_ratio	Total credit / average balance
withdrawal_to_balance_ratio	Total withdrawal / average balance
loan_to_credit_ratio	Loan payments / total credit
interbank_ratio	(Interbank in + Interbank out) / total transactions

4.2 Transaction Variability

Feature	Description
withdrawal_std_ratio	Balance std / total withdrawal
credit_std_ratio	Balance std / total credit

4.3 Product / Service Engagement Ratios

Feature	Description
insurance_ratio	Insurance payments / total transactions
loan_ratio	Loan payments / total transactions
household_ratio	Household payments / total transactions

4.4 Behavioral Flags

Feature	Description
is_active_customer	1 if total transactions > 100, else 0
relies_on_credit	1 if total credit > total withdrawal, else 0

4.5 Risk / Stability Indicators

Feature	Description
sanction_interest_ratio	Sanction interest payments / total transactions
volatility_index	Balance std / average balance

5. Final Base Table

- **Columns included:**
 - Client demographics (client_id, gender, age, age_group)
 - Account info (lor, frequency_eng)
 - Orders (num_orders_1996, total_order_amount_1996, order flags)
 - Transactions (total_credit, total_withdrawal, balance_mean, balance_std, counts, ratios)
 - RFM (frequency, monetary, recency)
 - District features (A2–A16)
 - Target variables at the **end**: loan_1997, card_1997
 - Missing values filled with 0.
 - All ratio features computed to avoid division by zero using `.replace(0,1)`.
 - `base_table.csv` created for further analysis or modeling.
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6. Usage

- `base_table` can be directly used for:
 - Exploratory data analysis
 - Feature correlation analysis
 - Modeling (classification for loans and credit cards)

Base Table Features with Data Types and Explanations

Here is a complete list of all columns in the base_table along with their data types and explanations:

Column Name	Data Type	Description
client_id	int64	Unique identifier for each client
gender	object	Client gender (M/F)
age	int64	Age of client in 1996
age_group	category	Age grouped by decade (e.g., 30, 40...)
lor	int64	Length of relationship with bank (years)
frequency_eng	object	Account transaction frequency (MONTHLY, WEEKLY, TRANSACTION)
num_orders_1996	int64	Number of payment orders in 1996
total_order_amount_1996	float64	Total amount of orders in 1996
has_insurance_order_1996	int64	Flag for insurance order in 1996 (0/1)
has_household_order_1996	int64	Flag for household order in 1996 (0/1)
has_leasing_order_1996	int64	Flag for leasing order in 1996 (0/1)
has_loan_repayment_order_1996	int64	Flag for loan repayment order in 1996 (0/1)
total_credit	float64	Total credit amount in 1996
total_withdrawal	float64	Total withdrawal amount in 1996
total_transactions	int64	Total number of transactions in 1996
balance_mean	float64	Average account balance in 1996
balance_std	float64	Standard deviation of balance

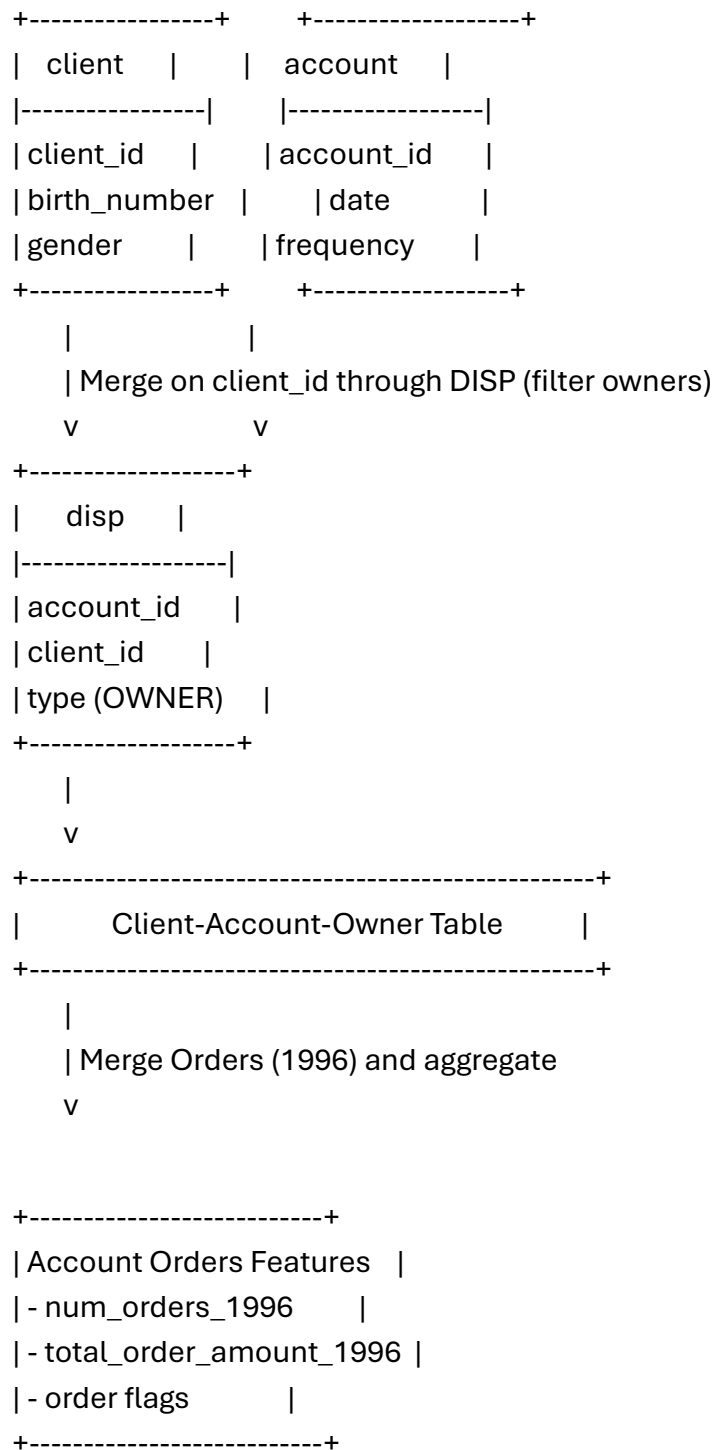
Column Name	Data Type	Description
credit_card_withdraw_count	int64	Number of credit card withdrawals
cash_deposit_count	int64	Number of cash deposits
interbank_in_count	int64	Number of incoming interbank transfers
interbank_out_count	int64	Number of outgoing interbank transfers
cash_withdrawal_count	int64	Number of cash withdrawals
insurance_payment_count	int64	Number of insurance payments
services_payment_count	int64	Number of service payments
interest_credit_count	int64	Number of interest credits
sanction_interest_count	int64	Number of sanction interest entries
household_payment_count	int64	Number of household payments
pension_credit_count	int64	Number of pension credits
loan_payment_count	int64	Number of loan payments
frequency	int64	Transaction frequency (RFM)
monetary	float64	Total monetary value of transactions (RFM)
recency	int64	Days since last transaction (RFM)
A2	object	District name
A3	object	Region name
A4	int64	Number of inhabitants in district
A5	int64	Municipalities with <499 inhabitants

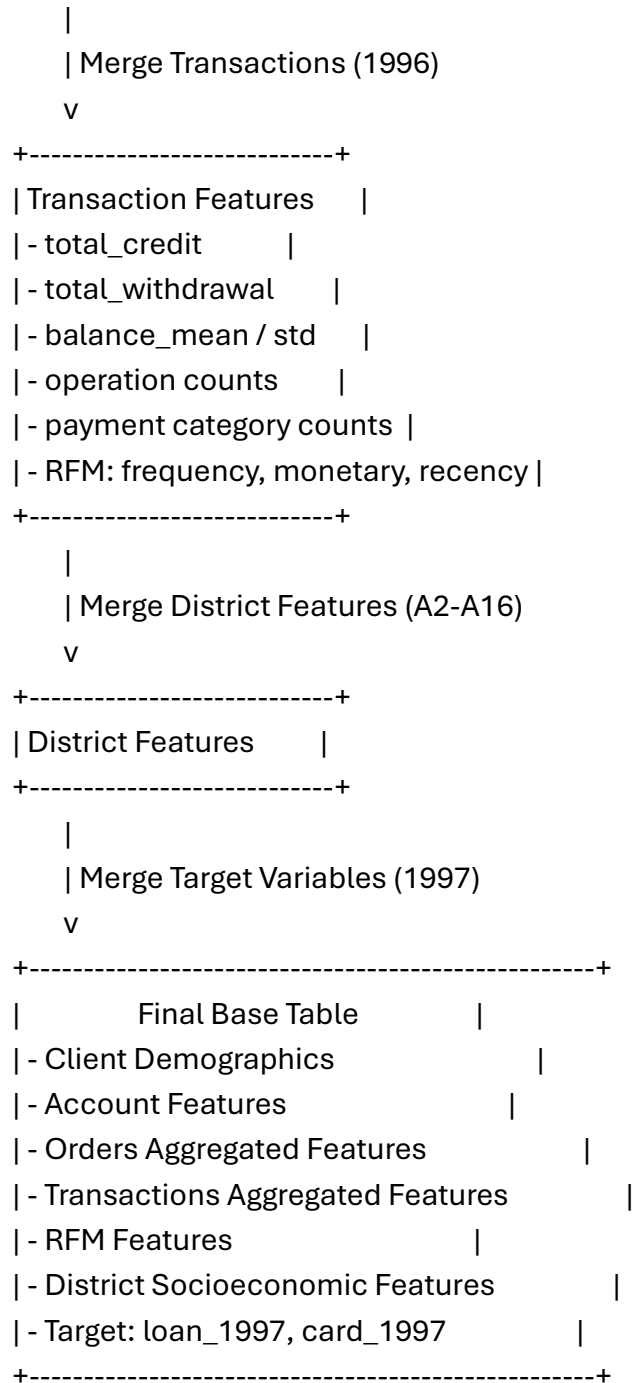
Column Name	Data Type	Description
A6	int64	Municipalities with 500–1999 inhabitants
A7	int64	Municipalities with 2000–9999 inhabitants
A8	int64	Municipalities with >10000 inhabitants
A9	int64	Number of cities in district
A10	float64	Ratio of urban inhabitants
A11	int64	Average salary in district
A12	float64	Unemployment rate 1995
A13	float64	Unemployment rate 1996
A14	int64	Entrepreneurs per 1000 inhabitants
A15	float64	Committed crimes 1995
A16	int64	Committed crimes 1996
loan_1997	int64	Target: 1 if client obtained loan in 1997, else 0
card_1997	int64	Target: 1 if client obtained credit card in 1997, else 0
credit_to_withdrawal_ratio	float64	Ratio of total credit to total withdrawal
net_cash_flow	float64	Total credit minus total withdrawal
net_cash_flow_ratio	float64	Net cash flow / total transactions
credit_to_balance_ratio	float64	Total credit / average balance
withdrawal_to_balance_ratio	float64	Total withdrawal / average balance
loan_to_credit_ratio	float64	Loan payments / total credit
interbank_ratio	float64	Interbank transfers / total transactions
withdrawal_std_ratio	float64	Balance std / total withdrawal

Column Name	Data Type	Description
credit_std_ratio	float64	Balance std / total credit
insurance_ratio	float64	Insurance payments / total transactions
loan_ratio	float64	Loan payments / total transactions
household_ratio	float64	Household payments / total transactions
is_active_customer	int64	Flag if total transactions > 100 (1=yes, 0=no)
relies_on_credit	int64	Flag if total credit > total withdrawal
sanction_interest_ratio	float64	Sanction interest / total transactions
volatility_index	float64	Balance std / mean balance (stability measure)

Base Table Creation Flowchart

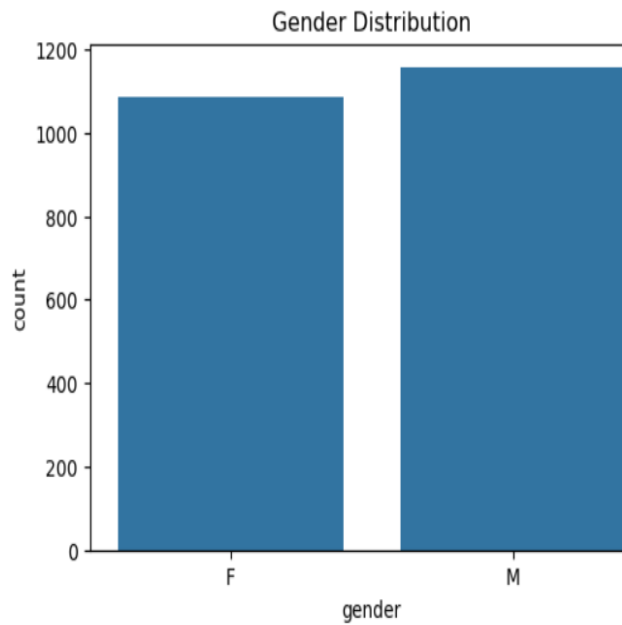
This diagram illustrates the process of creating the final base_table from multiple raw datasets, including feature engineering steps. The disp table is explicitly included to clarify the joining process.



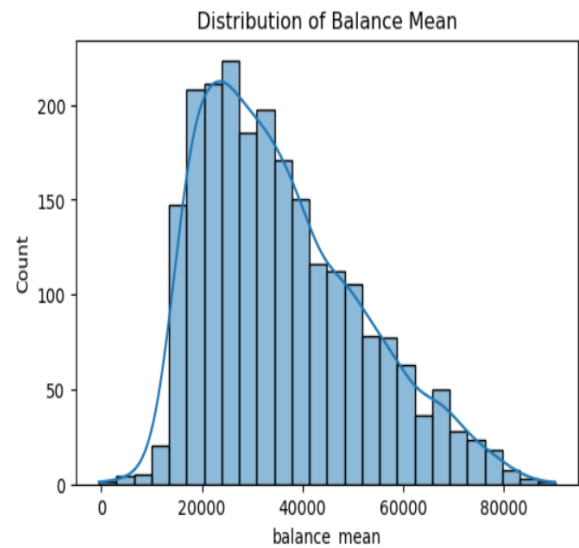


Notes: - Missing values are replaced with 0. - Ratio features are computed safely to avoid division by zero. - Behavioral flags and risk/stability indicators are included for modeling. - The base_table is ready for exploratory analysis and predictive modeling.

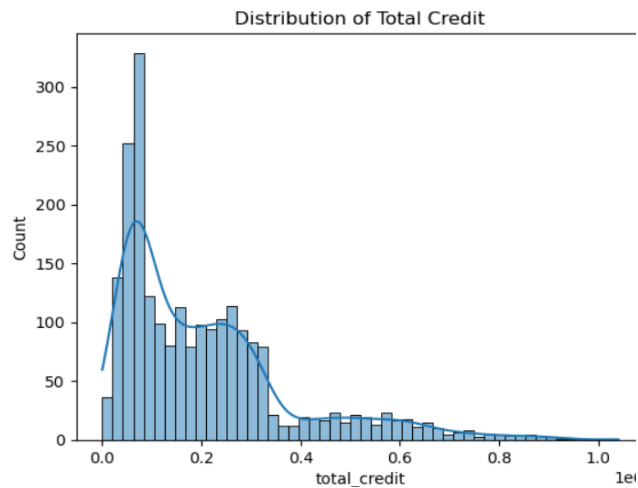
EDA EXPLORATION:



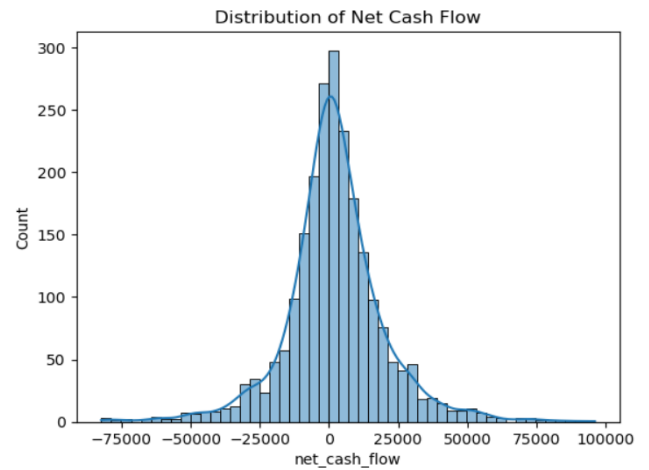
This bar chart displays the gender distribution within a dataset. The chart has two bars labeled "F" for Female and "M" for Male along the x-axis, while the y-axis shows the count of individuals for each gender. Both groups are almost equal, with each bar representing around 1,100–1,200 people.



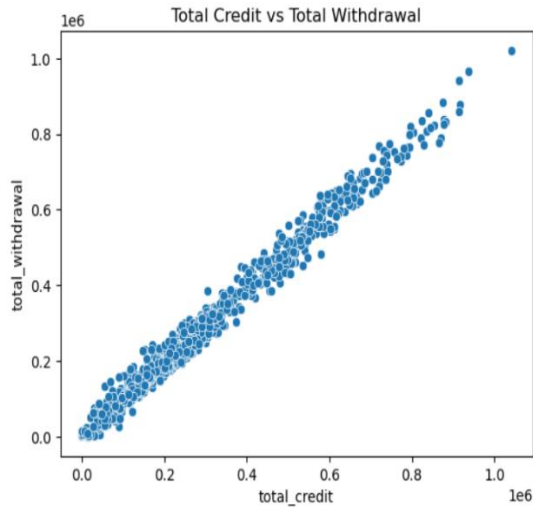
This chart shows the distribution of average account balances among bank customers in the base table. Most customers have balances between 10,000 and 40,000, with fewer having higher amounts. The data is right-skewed, indicating that high balances are less common.



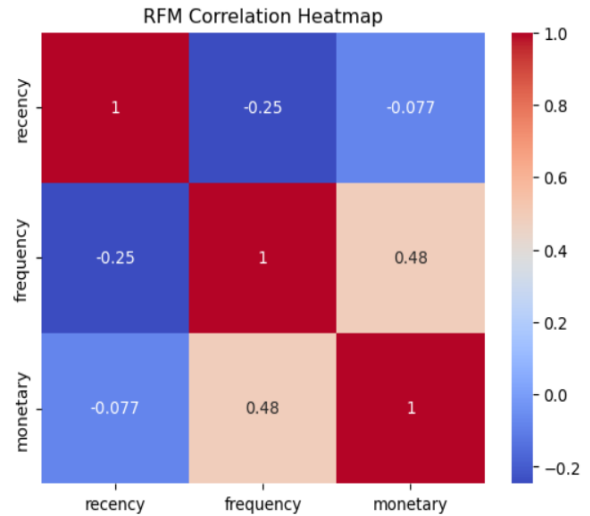
This chart shows the distribution of total credit among bank customers in the base table. Most customers have total credit below 200,000, with a long tail of fewer people having higher credit amounts. The distribution is heavily right-skewed, indicating high credit values are less frequent.



This chart displays the distribution of net cash flow among bank customers in the base table. The data is centered around zero, with most customers having cash flows close to zero and fewer at the extremes. The resulting pattern is symmetric and resembles a normal distribution.

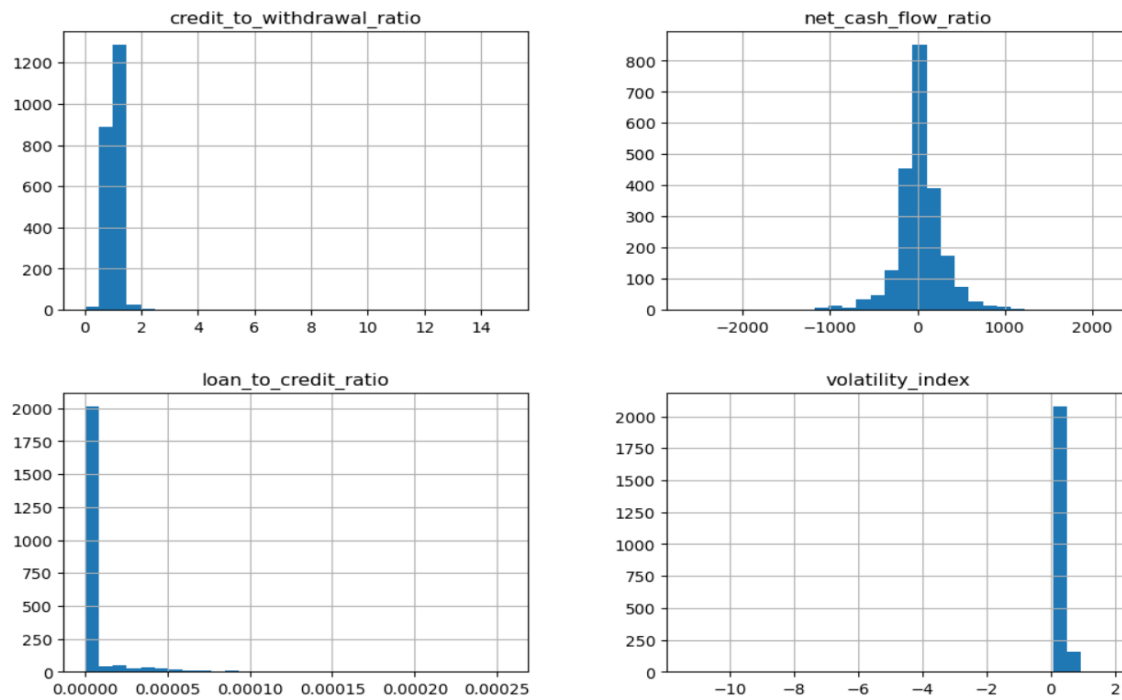


This scatter plot shows the relationship between total credit and total withdrawal for each customer in the dataset. The points form a strong diagonal pattern, indicating that customers who deposit (credit) more also tend to withdraw more. This suggests a strong positive correlation between total credit and total withdrawal amounts.



This heatmap shows the correlations between recency, frequency, and monetary values in the RFM (Recency, Frequency, Monetary) model. Recency and frequency have a negative correlation (-0.25), meaning that customers who purchased more recently tend to buy more frequently. Frequency and monetary have a positive correlation (0.48), indicating frequent buyers also spend more.

Distribution of Ratio Features



This figure displays the distribution of four ratio-based features for bank customers. Most values for `credit_to_withdrawal_ratio` and `loan_to_credit_ratio` are tightly concentrated near their lower ranges. The `net_cash_flow_ratio` is centered around zero, and `volatility_index` values are mostly clustered just below zero, indicating generally stable or low-variance account behavior.

Summary:

- **Total columns:** 65
- **Total rows:** 2239

Feature Groups:

1. **Client Demographics:** client_id, gender, age, age_group, lor, frequency_eng
2. **Orders (1996):** num_orders_1996, total_order_amount_1996, order flags
3. **Transactions (1996):** credit/withdrawal totals, balance stats, operation counts, payment category counts
4. **RFM Features:** frequency, monetary, recency
5. **District Socio-Economic Features:** A2-A16
6. **Engineered Features / Ratios:** credit_to_withdrawal_ratio, net_cash_flow, net_cash_flow_ratio, credit_to_balance_ratio, withdrawal_to_balance_ratio, loan_to_credit_ratio, interbank_ratio, withdrawal_std_ratio, credit_std_ratio, insurance_ratio, loan_ratio, household_ratio, sanction_interest_ratio, volatility_index, is_active_customer, relies_on_credit
7. **Target Variables (1997):** loan_1997, card_1997

Notes:

- Missing values are replaced with 0.
- Ratio features are computed safely to avoid division by zero.
- Behavioral and stability flags are included for predictive modeling.
- The base table is ready for exploratory data analysis and machine learning tasks.

Reference:

<https://github.com/sabirakhtar>